

ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 2 of 10

Properties, State, Process and Cycle

Intensive and extensive properties · specific properties · state and the two-property rule · process, path and cycle · point and path functions

Dr. Anantha Raman L., M.E., Ph.D. · Associate Professor, Department of Mechanical Engineering, Rajalakshmi Institute of Technology, Chennai

In this presentation

1. What a property is
2. Intensive and extensive properties
3. Specific properties
4. State and the two-property rule
5. Process and the named processes
6. Path and cycle
7. Point functions and path functions

1

Properties

Intensive, extensive and specific

What a property is

- Any measurable characteristic of a system whose value depends **only on the state** – not on how the state was reached
- Pressure p , temperature T , volume V , mass m , density ρ
- Internal energy U , enthalpy H , entropy S
- Properties are **point functions**: their differentials are exact

Intensive and extensive properties

Basis	Intensive	Extensive
Dependence on mass	Independent of the mass	Proportional to the mass
System divided in two halves	Value unchanged in each half	Value halved in each half
Examples	p, T, ρ, v, u, h	V, m, U, H, S
Notation	Lower-case letters (except p and T)	Capital letters

Specific properties

- Extensive property $\div$ mass = **specific** property, which is intensive
- Specific volume $v = V/m$ (m^3/kg)
- Specific internal energy $u = U/m$ (kJ/kg)
- Specific enthalpy $h = H/m$ (kJ/kg)

KEY POINT Quick test: imagine cutting the system in half – if the value changes, the property is extensive.

2

State

and the two-property rule

State of a system

- The complete condition of the system, identified by the values of its properties
- On a property diagram (p – V , T – s) a state is a **single point**
- **Simple compressible system** – a pure substance free of motion, gravity, electrical, magnetic and surface effects
- **Two-property rule** – two independent intensive properties fix the state; all others follow

KEY POINT Given p and T of superheated steam, or p and v of a gas, the state – and hence u , h , s – is fixed.

3

Process, path and cycle

Process

- The change of a system from one equilibrium state to another
- Named after the property that is held constant
- Processes are drawn as curves on property diagrams

The named processes

Process	What is constant	Example
Isobaric	Pressure	Heating water in an open pan; boiler tubes
Isochoric	Volume	Heating gas in a rigid sealed tank
Isothermal	Temperature	Very slow compression with perfect cooling
Adiabatic	No heat transfer, $Q = 0$	Rapid compression in an insulated cylinder
Isentropic	Entropy	Ideal (reversible adiabatic) turbine expansion

Path and cycle

- **Path** – the succession of states passed through: the curve joining the end points on a property diagram
- **Cycle** – a sequence of processes that begins and ends at the same state
- Over a cycle the net change of every property is zero: $\oint dp = 0$, $\oint dV = 0$, $\oint dU = 0$
- A process has definite, generally different, end states; a cycle returns to its starting point

4

Point functions and path functions

Point functions and path functions

Feature	Point function	Path function
Depends on	End states only	The particular path followed
Differential	Exact: dp, dV, dT, dU	Inexact: $\delta Q, \delta W$
Integrates to	$p_2 - p_1, U_2 - U_1$	Q_{1-2}, W_{1-2} – never $Q_2 - Q_1$
Cyclic integral	$\oint d(\text{property}) = 0$	$\oint \delta Q, \oint \delta W \neq 0$ in general
Examples	All properties: p, V, T, U, H, S	Heat Q and work W

Heat and work are not properties

A system *possesses* energy; it never "contains" heat or work. They exist only while crossing the boundary.

- Write $\Delta U = U_2 - U_1$, but never ΔQ
- Heat transferred in a process is written Q_{1-2} or simply Q

Summary

- Property: depends only on the state; exact differential
- Intensive – independent of mass (p , T , ρ , v , u , h); extensive – proportional to mass (V , U , H , S)
- Specific property = extensive $\div$ mass (v , u , h)
- State fixed by two independent intensive properties
- Process: isobaric, isochoric, isothermal, adiabatic, isentropic
- Cycle: returns to the initial state, $\oint d(\text{property}) = 0$
- Heat and work: path functions, δQ and δW inexact

Next: Equilibrium, quasi-static and reversible processes

- Mechanical, thermal and chemical equilibrium
- The quasi-static process and why it matters
- Reversible and irreversible processes; causes of irreversibility